DATA ANALYTICS PORTFOLIO & TECHNICAL DOCUMENTATION

Hotel Booking Analysis 2024

Tool: Microsoft Excel (Pivot Tables, Pivot Charts, Dynamic Slicers) | Dataset: hotel_bookings_updated_2024 (119,390 Records)
Data Source: **Kaggle — Hotel Booking Reservation (by Kundan Sagar Bedmutha)**

1. Project Objective

The primary objective of the **Hotel Booking Analysis 2024** project is to transform raw hospitality booking records (119,390 rows across 34 attributes) into an interactive, executive-ready Excel dashboard. By examining booking volumes, cancellation behaviors, booking lead times, customer segments, meal preferences, and geographical distributions across 15 major Indian cities, this dashboard empowers hotel management to identify revenue drivers, minimize high cancellation rates, and optimize seasonal pricing and operational planning.

> **Executive Highlight:** Across **119,390 total bookings** in 2024, the hotel chain experienced a **37.04% cancellation rate (44,224 cancellations)** with an average booking lead time of **104 days** and an Average Daily Rate (ADR) of **102**. Online Travel Agents (Online TA) drive nearly half (47.40%) of all reservations, and Bed & Breakfast (BB) accounts for 77.32% of guest meal choices.

2. Executive KPI Summary

Five top-level Key Performance Indicators (KPIs) are calculated on the dedicated `KPI's` worksheet and linked dynamically to the main dashboard header cards for rapid executive review:

KPI Metric	Dashboard Value	Exact Underlying Value	Business Meaning & Analytical Relevance
Total Bookings	**119k**	**119,390**	Total count of reservation records captured across all City and Resort hotels in 2024.
Cancelled Booking	**44k**	**44,224**	Total bookings flagged under `is_canceled`, representing lost or re-allocated room inventory.
Cancellation Rate	**37.04%**	**37.04%**	Percentage of total bookings cancelled; highlights a critical area for deposit and policy optimization.
Average Lead Time	**104**	**104 Days**	Average number of days between the booking date and actual arrival date across all guests.
Average ADR	**102**	**102**	Average Daily Rate (revenue earned per occupied room), formatted to whole numbers for clarity.

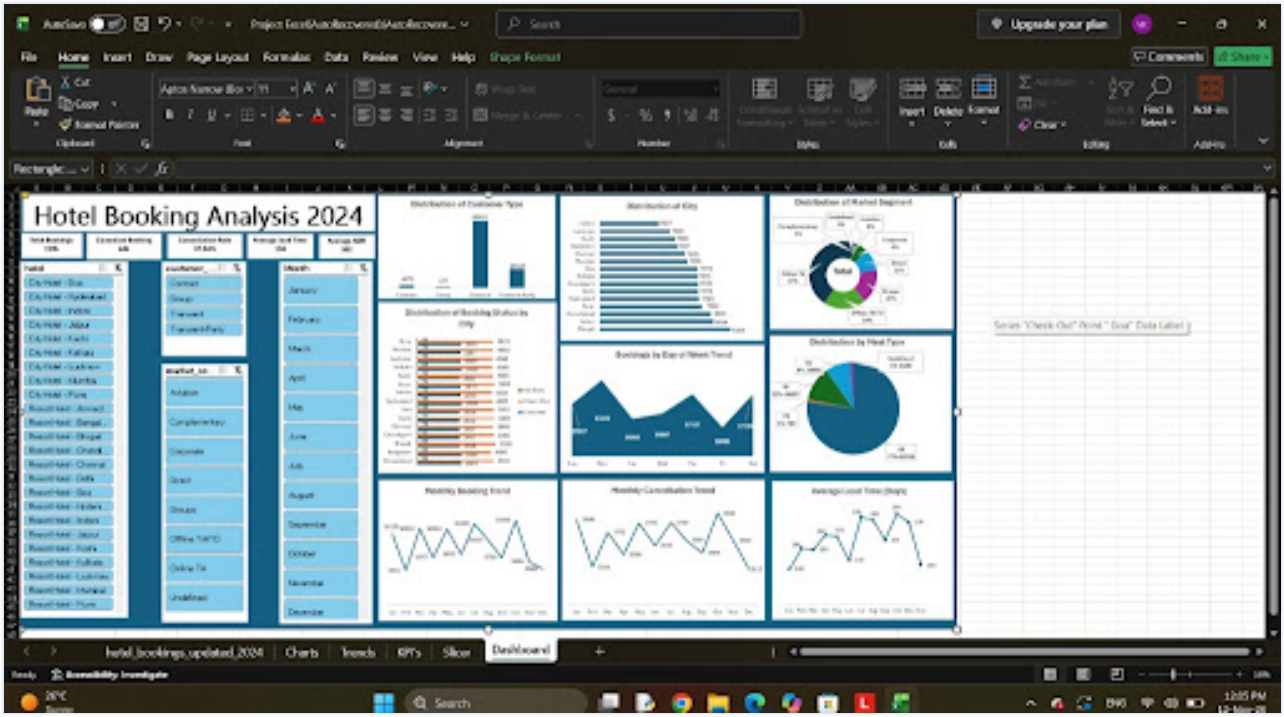


Figure 1: Complete Interactive Hotel Booking Analysis 2024 Dashboard in Microsoft Excel

3. Dataset Architecture & Data Preparation

The base data is sourced from the **Kaggle Hotel Booking Reservation Dataset** (`hotel_bookings_updated_2024.csv`, License: CC BY 4.0) and housed in the `hotel_bookings_updated_2024` sheet as a structured Excel Table spanning Columns A through AH (34 columns). Formatting the dataset as an official Excel Table ensures that all downstream Pivot Tables dynamically reference accurate ranges.

Category	Dataset Columns (Attributes)	Purpose in Analysis
Property & Geography	`hotel`, `City`, `country`	Identifies property type (City Hotel vs. Resort Hotel) across 15 Indian cities (Bhopal, Jaipur, Ahmedabad, Pune, etc.).
Booking Status & Timing	`is_canceled`, `lead_time`, `reservation_status`, `reservation_status_date`, `Day`, `Month`, `Year`, `Week number`, `Week day`	Tracks Check-Out, Canceled, and No-Show outcomes, advance booking window, and temporal seasonality across 2024.
Guest & Stay Profile	`stays_in_weekend_nights`, `stays_in_week_nights`, `adults`, `children`, `babies`, `meal`, `customer_type`, `is_repeated_guest`	Evaluates stay duration, party composition, meal package selection (BB, HB, SC, FB), and guest classification.
Commercial & Channel	`market_segment`, `distribution_channel`, `adr`, `deposit_type`, `agent`, `company`	Measures acquisition channels (Online TA, Offline TA/TO, Groups, Direct, Corporate) and room pricing (ADR).

Category	Dataset Columns (Attributes)	Purpose in Analysis
Operational Details	`previous_cancellations`, `previous_bookings_not_canceled`, `reserved_room_type`, `assigned_room_type`, `booking_changes`, `days_in_waiting_list`, `required_car_parking_spaces`, `total_of_special_requests`	Captures operational modifications, room upgrades, waiting list durations, and special guest requirements.

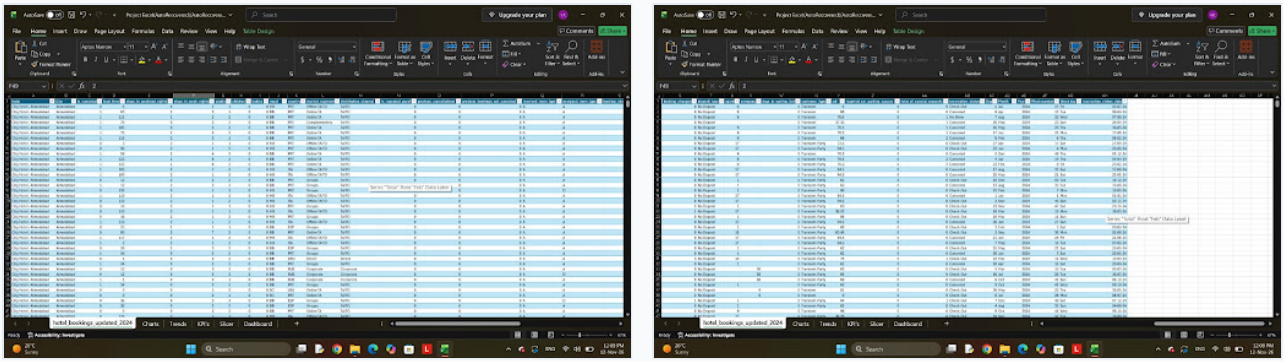


Figure 2: Structured Source Dataset (hotel_bookings_updated_2024) — Columns A to S (Left) and Columns S to AH (Right)

4. Analytical Breakdown: Charts Worksheet

The `Charts` worksheet contains five core Pivot Tables and their corresponding visualizations, examining customer demographics, acquisition channels, meal preferences, and city-level performance:

Analysis Dimension	Category / Segment Breakdown	Volume / Share	Key Analytical Finding
1. Customer Type *(Column Chart)*	Transient Transient-Party Contract Group	**89,613 (75.06%)** **25,124 (21.04%)** **4,076 (3.41%)** **577 (0.48%)**	Individual **Transient** guests account for three-quarters of all bookings, followed by Transient-Party bookings.
2. Market Segment *(Doughnut Chart with Center "Total" Label)*	Online TA Offline TA/TO Groups Direct Corporate Complementary Aviation / Undefined	**47.40% (~47%)** **20.33% (~20%)** **16.63% (~17%)** **10.56% (~11%)** **4.44% (~4%)** **0.62% (~1%)** **<0.20% (0%)**	**Online Travel Agents (47.40%)** and **Offline TA/TO (20.33%)** together generate over 67% of total hotel demand.

Analysis Dimension	Category / Segment Breakdown	Volume / Share	Key Analytical Finding
3. Meal Type *(Pie Chart with Count & Percentage Labels)*	BB (Bed & Breakfast) HB (Half Board) SC (Self Catering) Undefined FB (Full Board)	**92,310 (77.32%)** **14,463 (12.11%)** **10,650 (8.92%)** **1,169 (0.98%)** **798 (0.67%)**	**BB (77.32%)** is overwhelmingly preferred by guests, whereas Full Board (FB) represents less than 1% of bookings.
4. City Distribution *(Horizontal Bar Chart)*	Top 3: Bhopal (8,104), Jaipur (8,038), Ahmedabad (8,028) Mid-tier: Pune (7,992), Hyderabad (7,981), Delhi (7,978), Chandigarh (7,978), Kolkata (7,976), Goa (7,973), Mumbai (7,935), Chennai (7,925) Lowest 4: Bangalore (7,897), Kochi (7,889), Lucknow (7,869), Indore (7,827)	**119,390 Total (~7,827 to 8,104 per city)**	Booking volume is evenly distributed across all 15 cities, with **Bhopal (8,104)** leading and **Indore (7,827)** at the lower bound.
5. Booking Status by City *(Clustered Horizontal Bar Chart)*	Check-Out (Completed Stays) Canceled No-Show	**75,166 (62.96%)** **43,017 (36.03%)** **1,207 (1.01%)**	Across every city, roughly ~5,000 bookings result in Check-Out while ~2,800–2,900 are Canceled (e.g., Bhopal: 5,182 Check-Out, 2,836 Canceled, 86 No-Show).

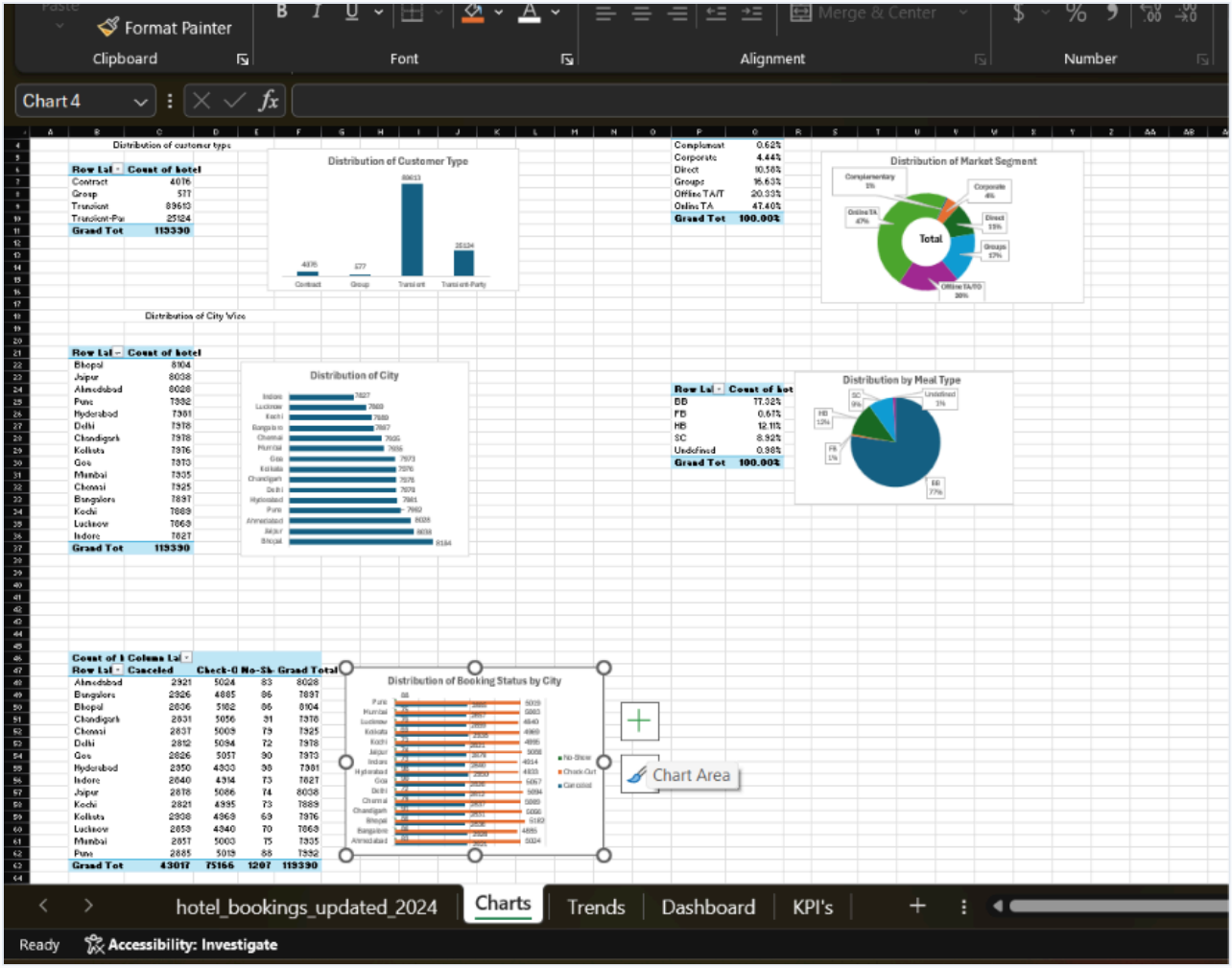


Figure 3: Charts Worksheet — Customer Type, Market Segment, City Distribution, Meal Type, and City Booking Status

5. Temporal & Behavioral Trends: Trends Worksheet

The `Trends` worksheet tracks monthly and day-of-week patterns across four dedicated Pivot Tables and line/area visualizations to uncover seasonality in bookings, cancellations, and advance planning:

Month	Total Bookings (Count of hotel)	Cancelled Bookings (Sum of is_canceled)	Average Lead Time (Days)
January	10,126	3,808	102
February	9,551	3,524	104
March	10,216	3,732	103
April	9,757	3,598	104
May	10,212	3,788	104
June	9,811	3,653	103
July	10,298	3,789	105
August	10,033	3,685	105
September	9,759	3,609	104

Month	Total Bookings (Count of hotel)	Cancelled Bookings (Sum of is_canceled)	Average Lead Time (Days)
October	**10,349**	**3,846**	**105**
November	**9,696**	**3,680**	**105**
December	**9,582**	**3,512**	**103**
Grand Total / Overall Average	**119,390**	**44,224**	**104**

Bookings by Day of Week Trend (Area Chart): Evaluating weekly arrival and status patterns shows consistent high demand across all seven days, peaking on **Monday (17,221)**, **Saturday (17,128)**, and **Thursday (17,127)**, followed by **Sunday (17,057)**, **Wednesday (16,997)**, **Tuesday (16,959)**, and **Friday (16,901)**.

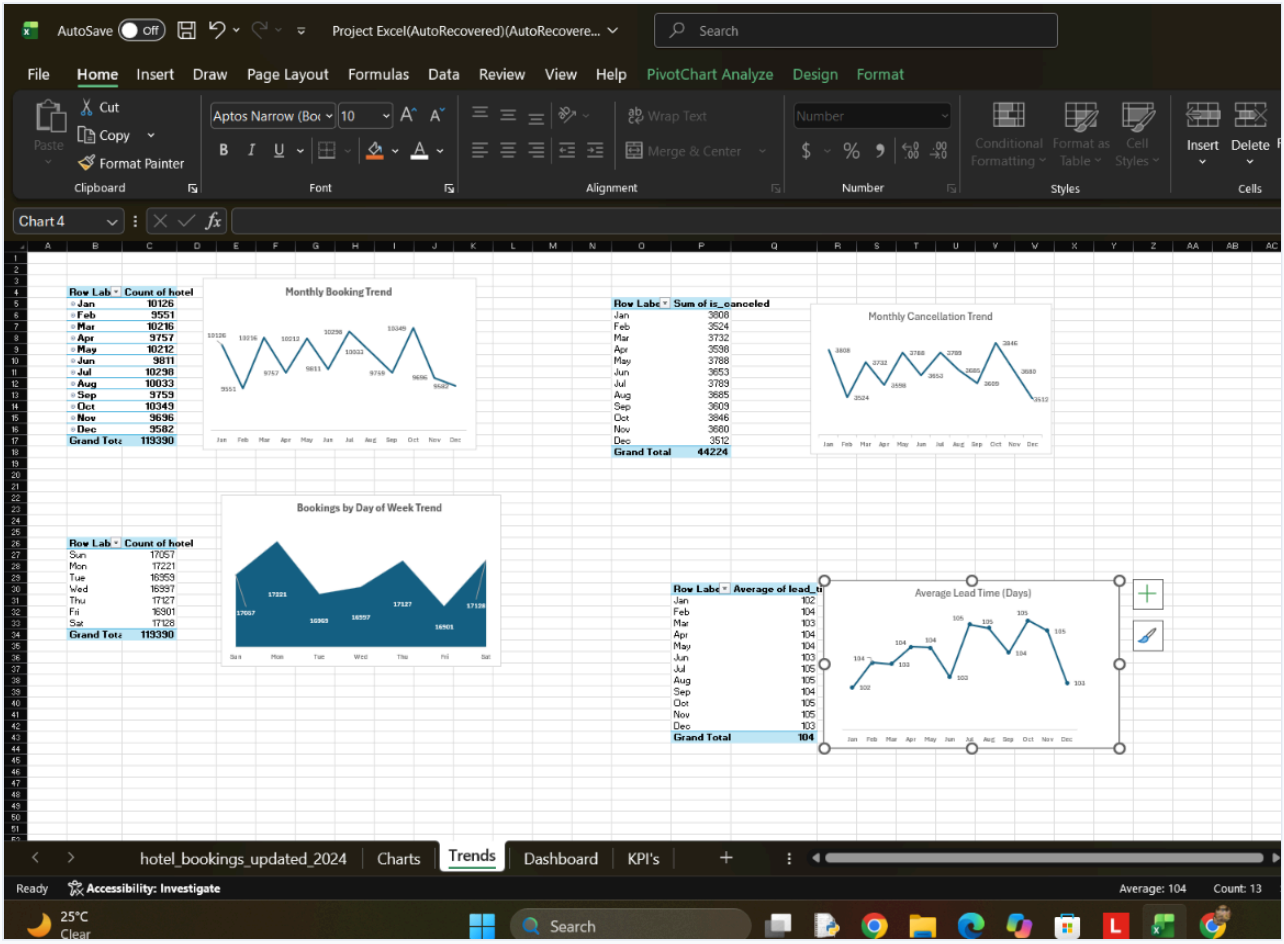


Figure 4: Trends Worksheet — Monthly Booking Trend, Monthly Cancellation Trend, Day of Week Area Chart, and Average Lead Time

6. Interactive Slicers & Dashboard Architecture

To make the dashboard fully interactive, four custom-styled Slicers were created on the `Slicer` worksheet and integrated into the left navigation panel of the dark-blue `Dashboard` canvas:

- **`hotel` Slicer:** Filters the entire dashboard by specific property and city (e.g., *City Hotel - Ahmedabad* through *Resort Hotel - Pune*).
- **`customer_type` Slicer:** Segments metrics across *Contract*, *Group*, *Transient*, and *Transient-Party* guests.
- **`market_segment` Slicer:** Isolates performance for *Aviation*, *Complementary*, *Corporate*, *Direct*, *Groups*, *Offline TA/TO*, *Online TA*, and *Undefined* channels.
- **`Month` Slicer:** Enables single-month or multi-month seasonal filtering from *January* to *December*.
- **Report Connections Integration:** Each slicer is linked via *Right-Click → Report Connections* to all 9 Pivot Tables across the `Charts` and `Trends` sheets so that every chart updates simultaneously upon user selection.

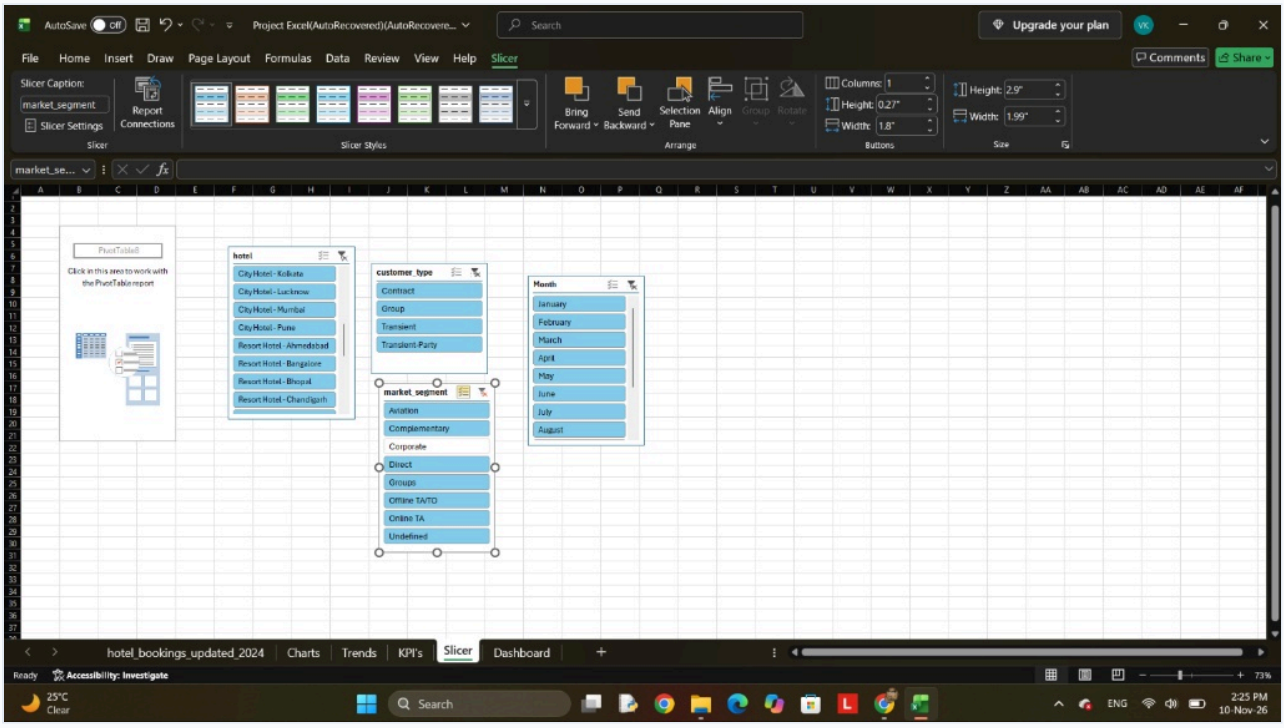


Figure 5: Slicer Configuration Sheet Featuring Hotel, Customer Type, Market Segment, and Month Filters

7. Key Business Recommendations & GitHub Portfolio Checklist

- **Mitigate Online TA Cancellations:** With a 37.04% overall cancellation rate and 47.40% of bookings originating from Online Travel Agents, management should introduce tiered non-refundable discounts or partial deposits for bookings made more than 100 days in advance.
- **Peak Month Inventory Management:** October (10,349 bookings; 3,846 cancellations) and July (10,298 bookings; 3,789 cancellations) experience the highest volume and cancellation spikes—requiring strategic overbooking buffers.
- **Meal Package Upselling:** Since 77.32% of guests select Bed & Breakfast (BB) and 8.92% opt for Self Catering (SC), discounted check-in upgrades to Half Board (HB) can boost ancillary food and beverage revenue.
- **GitHub Repository Structure:** Upload the final saved workbook (`Hotel_Booking_Analysis_2024.xlsx`), dashboard screenshot, this PDF documentation report, and cite the official Kaggle dataset link (**https://www.kaggle.com/datasets/kundanbedmutha/hotel-booking-reservation**) in your GitHub repository README.